

Question				Marking details		Marks available					
						AO1	AO2	AO3	Total	Maths	Prac
1	(a)			Name	Location	3			3		
				glycolysis	Cytoplasm/cytosol (1)						
				Link (reaction)	Matrix (of mitochondrion) (1)						
				Krebs (cycle)	Matrix (of mitochondrion) (1)						
				1 mark for each row							
	(b)			Decarboxylase (1) (Released as) <u>CO₂</u> (1)			2		2		
	(c)	(i)		Reduction/ hydrogen gain (1)			1		1		
		(ii)	I	black one is NAD grey one is FAD (1) Reject NADP			1		1		
			II	Any three (×1) from: A. {Transport/ donate} {H atoms/electrons/ H ⁺ / protons} to {ETC / inner mitochondrial membrane/ electron carriers} (1) B. NAD delivers to first proton pump and FAD to second /NAD uses 3 (types of) proton pumps and FAD uses 2 (1) C. (Reduced) NAD results in more protons being pumped (across membrane) than reduced FAD /Reduced NAD leads to a steeper proton gradient. (1) D. NAD gives 3ATP FAD gives 2ATP (1) E. NAD involved in glycolysis + link + Krebs but FAD only Krebs (1)			3		3		
		(iii)		<u>Final</u> {Hydrogen/electron} acceptor (1)		1			1		
				Question 1 total		4	7	0	11	0	0